

Feed the future:How can farm households combat climate change in Bangladesh?

Development Economics @ Chiba University

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Self introduction

- Mar 2020 : Department of Food and Environmental Economics, Kyoto U.
 - Aug 2017~June 2018 : Exchange program University of the Philippines, Los Baños
- Jan 2022 : M.Sc. (Agric. Econ.), National Taiwan University
- Apr 2022 : Mitsubishi UFJ Research & Consulting Ltd.
- Nov 2022 : IDE-JETRO
- Oct 2024 : National Graduate Institute for Policy Studies (GRIPS), Policy Analysis program

Self-introduction

- Main interests : Agricultural Economics, Development Economics
 - Climate change adaptation, food security, child health, gender inequality in Bangladesh
 - Area studies on Bangladeshi politics and economy like CIA's Global Trend Report

IDEスクエア



ハシナ政権の崩壊——バングラデシュの政治・経済はどこに向かうのか

IDE-JETRO



IDEスクエア

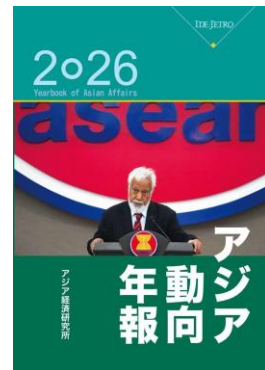


過渡期のバングラデシュ——七月政変から第13回総選挙に向かって

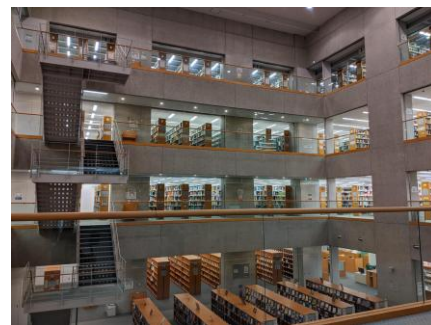
Introduction to IDE-JETRO in Kaihin Makuhari

- Research institute for social science on developing countries under METI
 - **Academic excellence** : Publishing academic papers
 - **Social commonality** : Publishing policy brief, outreach activities such as lecture outside academia
- Check out the web magazine of IDE

IDE-JETRO



おしえて! 知りたい!
途上国とSDGs



Today's Topic

- How can poor farm households in developing country enjoy sufficient and balanced diet and nutrition.
 - Outlook of household food and nutritional security
 - Barriers to household food and nutritional security
 - My recent research on food and nutritional security
 - Invitation to agricultural and development economics research



ORIGINAL ARTICLE

Weather shocks, livelihood diversification, and household food security: Empirical evidence from rural Bangladesh

Masanori Matsuura, Yir-Hueih Luh, Abu Hayat Md. Saiful Islam

First published: 15 May 2023 | <https://doi.org/10.1111/agec.12776> | [VIEW METRICS](#)

IDE-JETRO



Food Policy
Volume 135, August 2025, 102918



Weather shocks and child nutritional status in rural Bangladesh: Does labor allocation have a role to play?

Kirara Homma^a, Abu Hayat Md. Saiful Islam^b,
Masanori Matsuura^{c,d}, Bethelhem Legesse Debela^e

Concept of Household Food Security

Main Theme

How can vulnerable farmers achieve a sufficient and nutritional diet?

Two Dimensions

- **Food Accessibility:** Purchasing power measured by per capita food expenditure.
- **Food Utilization:** Ability to afford a nutritionally balanced diet (Dietary Diversity Index).

Note

Often framed as "Economic Security," focusing on macro risks like trade and national production. Today, we focus on the **micro-level perspective**.

Household Food Security: Global Comparison of Food Expenditure

United States

~\$3,700 per person / year

Japan

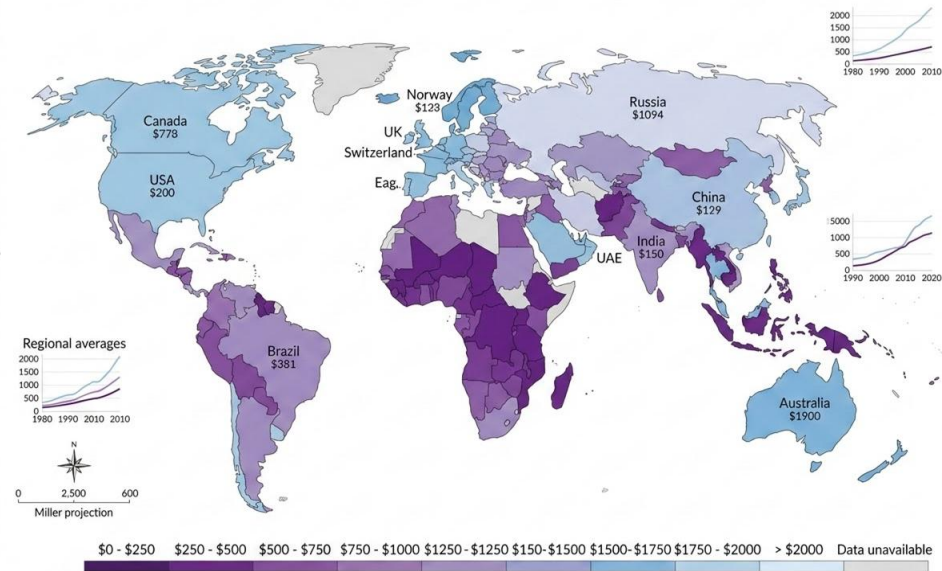
~\$2,800 per person / year

Bangladesh

~\$900 per person / year

ANNUAL FOOD EXPENDITURE PER PERSON BY COUNTRY

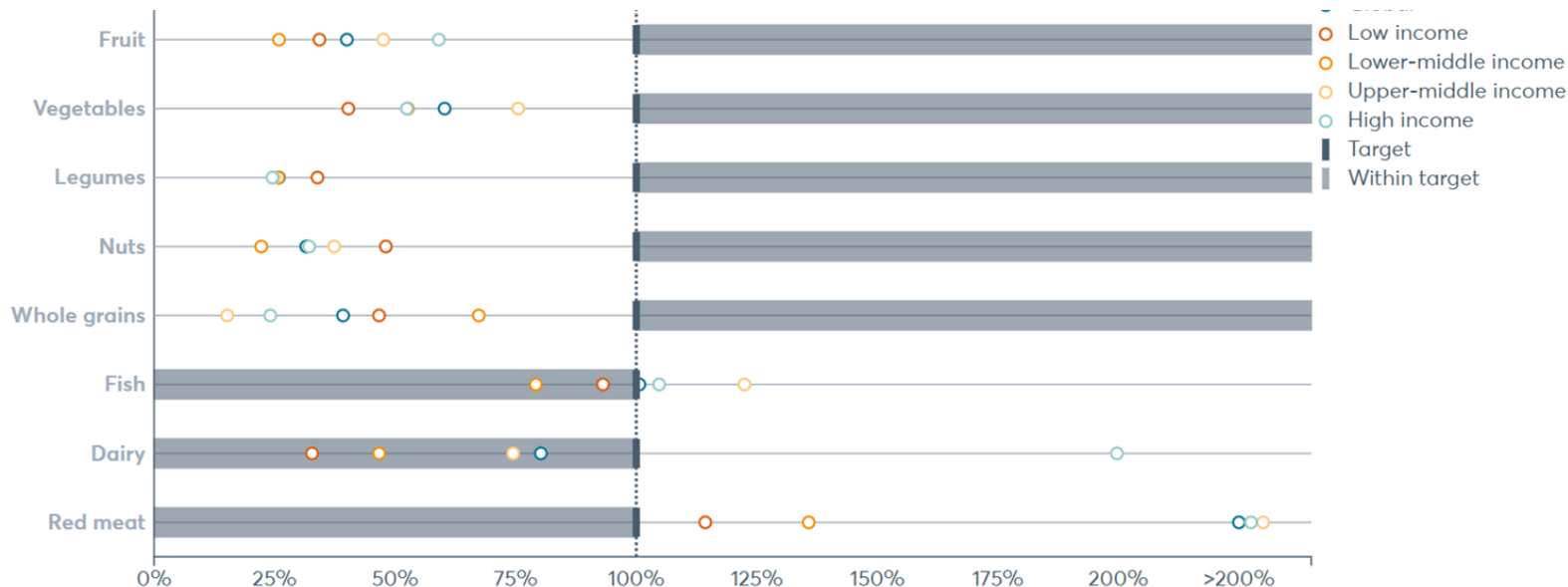
Estimates based on household surveys and national accounts (current US\$ per capita)



Source: Example based on data from World Bank, FAO, and national sources. For illustrative purposes.

Our Wo Miller 2023

Household Food Security: Dietary Diversity (Ages 20+)



Key Regional Insights

- High & Upper-Middle Income: High protein intake; relatively high fruit and vegetable consumption.
- Low & Lower-Middle Income: High legume and grain intake; insufficient protein, fruit, and vegetable consumption.

Household Food Security: Consequences of Inadequate Intake

Impact on Health & Nutrition

Leads to Malnutrition:

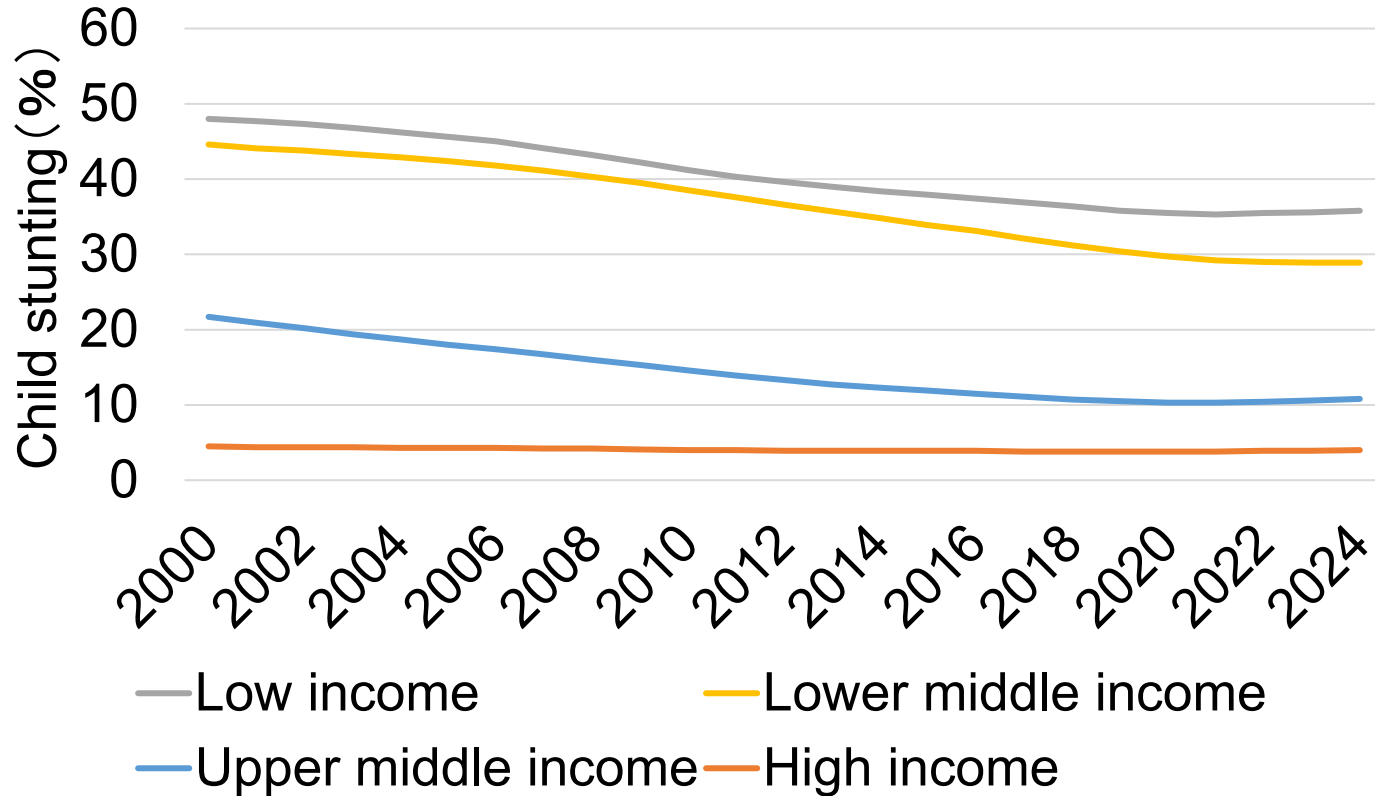
- **Child Stunting:** (0-59 months) Low height-for-age
- **Child Wasting:** Low weight-for-age
- **Anemia** and **Low Birth Weight**
- **Childhood Overweight:** Rising concern in high-income countries
- **Raised Blood Pressure**

Economic Development

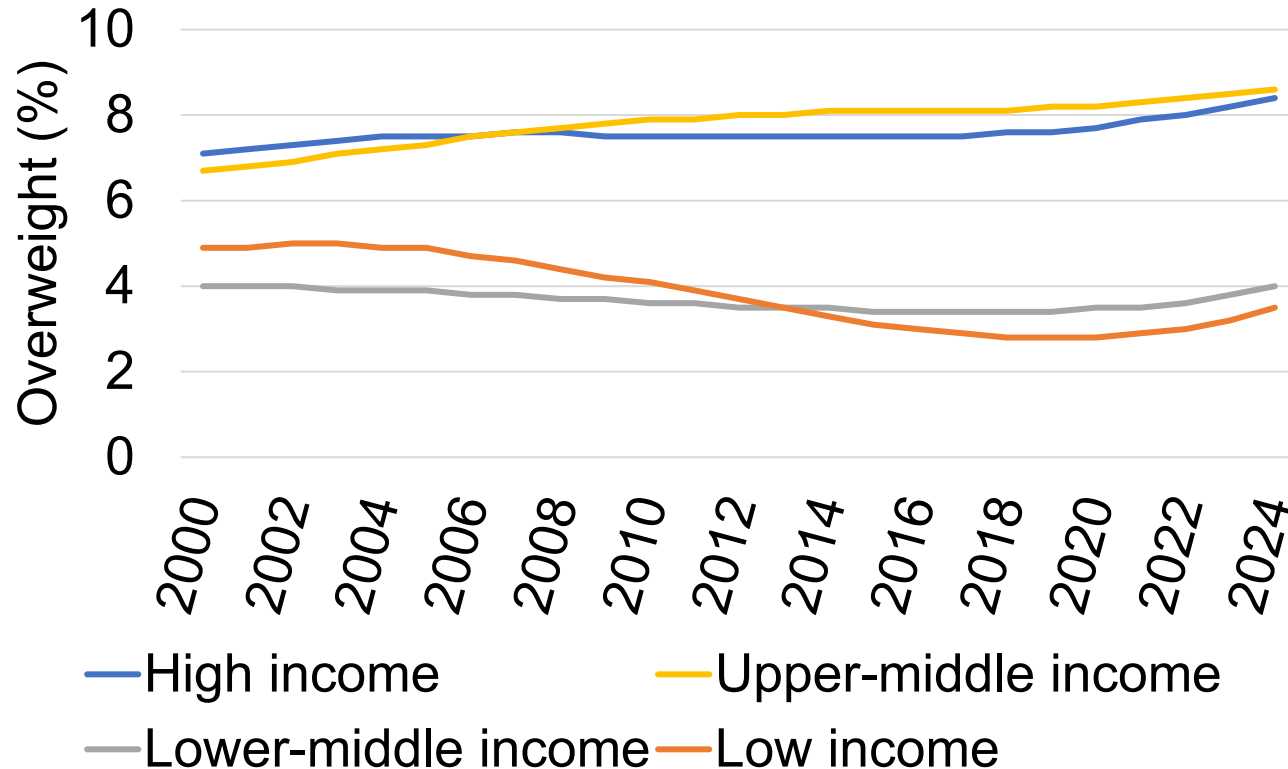
Why Nutrition Status Matters:

- **Labor Productivity:** Direct link to economic output

Household Food Security: Child Stunting



Household Food Security: Overweight



Barriers to Food Security: Challenges & Solutions

Climate Risks: Production Side

- Utilize high-yield varieties
- Adopt flood, drought, and pest-tolerant varieties
- Improve irrigation systems & soil quality
- **Diversify crops** (Avoid monoculture)
- **Increase non-agricultural income**

Disease Risks: Livestock

- One Health Approach: Simultaneously protect animal, environmental, and human health



Barriers to Food Security: Challenges & Solutions

Post-Harvest Spoilage of Crops

- Improving hygiene in value chains
- Addressing the difficulty of food preservation in developing countries

Nutrient-Poor Varieties & Food Safety

- Development of nutrient-rich varieties
- Removal of bacteria and parasites

Underdeveloped Transportation Infrastructure: Reducing transaction cost

- Improving market access and fixing road networks
- Bridging information asymmetry between farmers and brokers
- Expansion of IT technology (Smartphones, Feature phones)

My Works on Bangladesh

- Matsuura, M., Luh, Y. H., & Islam, A. H. M. S. (2023). Weather shocks, livelihood diversification, and household food security: Empirical evidence from rural Bangladesh. *Agricultural Economics*, 54(4), 455-470.
- Homma, K., Islam, A. H. M. S., Matsuura, M., & Debela, B. L. (2025). Weather shocks and child nutritional status in rural Bangladesh: Does labor allocation have a role to play?. *Food Policy*, 135, 102918.



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Current Situation in Bangladesh

Climate Vulnerability & Hazards

- Sea-level rise and cyclones causing flooding and casualties
- Economic damage and loss of life from extreme weather

Food Security & Health

- Heat stress leading to low agricultural productivity
- Rising temperatures increasing risks of cholera and heatstroke

Gender-Specific Risks

- Higher drowning risk for women during floods
- Increased risk of gender-based violence due to climate stress

Agricultural Sector Suffers Climate Change

Dry Season in Bangladesh



Flooding in Bangladesh



Mosquito Net in Bangladesh



How do Bangladeshi Farmers Adapt?

Livelihood Diversification

Addressing agricultural production risks through:

- *Crop diversification?*
- *Increasing non-farm income?*

Approached via **income diversification** and **labor allocation**.

Key Outcomes of Interest

- Food consumption per capita
- Household Dietary Diversity Score (HDDS)
- Child stunting

Research Question

Does income diversification and labor reallocation improve food security and child health?

Can these serve as effective adaptation strategies against weather shocks?

Farmers with Raising Livestock



Methodology: how to answer RQ

Micro-level Household Data

Utilizing Representative Datasets:

- Bangladesh Integrated Household Survey (2011/12, 2015, 2018/19)
- Collected by IFPRI (International Food Policy Research Institute)

Other Open Micro-data:

- World Bank: Living Standard Measurement Survey
- USAID: Demographic and Health Survey

Data Integration & Climate

Combining Weather and Survey Data:

- Merging climate records with household-level economic variables to identify correlations.

Measuring Weather Shocks:

- Utilizing rainfall and temperature data measured by the Bangladesh Meteorological Department.

Methodology: How to Answer RQ

Causal Inference

Technical Details (Econometrics):

- IV, DID, RDD, and RCT

Machine Learning Applications:

- Random forest
- T-learners, S-learners etc.

Qualitative Insight

Farmer Interviews:

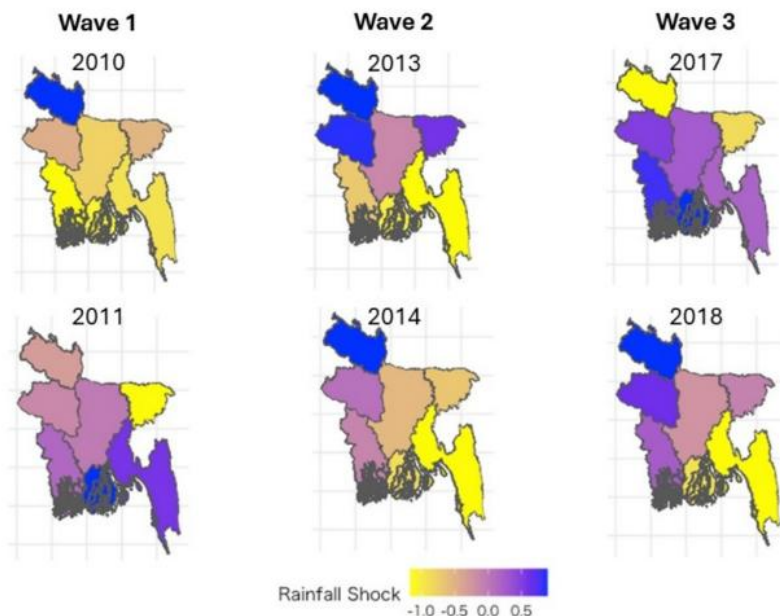
Interviews with local farmers in Bangladesh.

Objective:

- Understand local socioeconomic context
- Assist in interpreting complex statistical results

Why Do We Need Statistical Analysis?

- Various economic and geographic condition across divisions
- We need to account for those factors



Snapshot of My Interview in Bangladesh



Coexistence of Rice Farming and Shrimp Farming



Does Livelihood Diversification Improve Food Security?

Partially Yes

Key Findings

- **Crop diversification** does not significantly increase per capita food expenditure or dietary diversity.
- **Income diversification** significantly increases food security by securing sources resilient to climate risks.

The "Wealth Gap"

- Benefits are concentrated in **wealthier** farm households.
- Poor, low-education households in peripheral areas struggle to access quality non-farm labor markets.

Policy Implications

- Diversification policies alone are **insufficient** for the poorest farmers.
- Must simultaneously prioritize **high-quality job creation** in the non-agricultural sector.

Can Labor Reallocation Mitigate Negative Impacts of Weather Shocks?

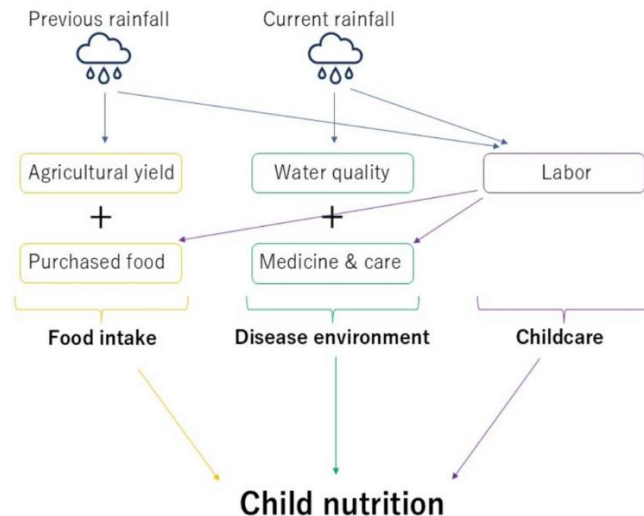
Impact of Weather Shocks on Child Health

1. Previous Year Drought

- Low rainfall in the previous year worsens child development.
- **Mechanism:** Decreased agricultural yield reduces food availability for the current year.

2. Current Year Flooding

- Excessive rainfall (flooding) in the current year stunts child growth.
- **Mechanism:** Water quality degradation in areas lacking clean sanitation leads to disease.



Can Labor Reallocation Mitigate Negative Impacts of Weather Shocks?

Baseline Condition (No Shock)

Children of mothers spending more time in agriculture have better health.

Reason: Non-farm work often requires being away from home, reducing childcare time.

Family-Wide Shocks

Time allocation changes in response to shocks don't significantly affect child health.

Factor: Mothers continue to bear the primary burden of care within the household.

Response to Little Rainfall

Mothers increasing non-farm self-employment can mitigate health deterioration.

Reason: Non-farm income improves access to food and medicine.

Policy Implications & Barriers

Policies enabling mothers to earn non-farm income are essential.

Challenge: Social norms against women working outside in some Muslim nations.

Conclusion: Invitation to Development Economics

Key Research Findings

Health impacts of poor diet are severe in developing nations, yet high-income countries suffer as well.

Barriers to achieving global food security remain multi-faceted and complex.

Empirical Research on Bangladesh

Estimated the effects of climate change adaptation behaviors in rural Bangladesh.

Demonstrates the useful case of empirical analysis for policy implications

Message to Students

If you have the chance, please pursue data analysis on a topic that interests you for your thesis.

Even without thesis, the ability to objectively interpret data is a vital skill in any jobs.

Broadening Horizons

Not necessarily "Go to Bangladesh," but expose yourself to diverse environments.

Contact Information

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Inquiries & Consultation

Please feel free to reach out regarding the following topics:

- Bangladesh
- Agricultural & Development Economics
- Study Abroad / International Exchange

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